Weekly Schedule

Spring 2027 · Mondays are lecture, Wednesdays are tutorials

Mondays are lecture; **Wednesdays** are hands-on tutorials that feed into that week's assignment. Data Safari exercises appear in Weeks 4, 7, 10, and 13. Slides, tutorials, and downloadable assignment templates are on the course website.

Week	Topic	Assignment	Data Safari	Note
1	Why experimental design matters; setting up R, RStudio, and Quarto	PS1		No lecture on Monday
2	R fundamentals and data wrangling	PS2		
3	Exploring data I: distributions and summary statistics	PS3		
4	Exploring data II: relationships and group comparisons	PS4	Safari 1	
5	Principles of experimental design: replication, randomization, blocking	PS5		Monday Holiday - Monday Schedule on Tuesday the 16th
6	Design in practice: sketching designs from the published literature	PS6		
7	Hypothesis testing I: the t-test and one-way ANOVA	PS7	Safari 2	
8	**Spring Break**			
9	Everything is a linear model: t-tests, ANOVA, and correlation as `lm()`	PS9		
10	Multiple regression, ANCOVA, and interactions; marginal effects	PS10	Safari 3	

Week	Topic	Assignment	Data Safari	Note
11	Generalized linear models: binary and count data; chi-square as a linear model	PS11		
12	Non-independence: mixed-effects models I (random intercepts)	PS12		
13	Mixed-effects models II: random slopes, nested vs. crossed designs	PS13	Safari 4	
14	When assumptions fail (ranks, robust SEs, permutation) and choosing a model	PS14		Monday Holiday - Monday Schedule on Wednesday the 21st
15	**Final Project Presentations**			

Graduate project milestones (BIOL 5860 only)

Four checkpoint deliverables spread across the semester, each documenting one stage of your independent project. See the Assignments overview for the full description. As with weekly assignments, click the name to download the blank `.qmd` template.

Due by end of	Milestone	Template
Week 7	Experimental design and planned hypothesis testing	Part 1
Week 10	Data carpentry and exploration	Part 2
Week 13	Statistical modeling	Part 3
Week 14	Visualization	Part 4
Week 15	Final presentation covering the entire process	—